

MIT OpenCourseWare

2.002

Mechanics and Materials II 2.002 — Mechanics and Materials II May 2026

EXTREMELY HARD – MIT LEVEL**Time Limit: 3 hours****Total Marks: 100**

1. (5 points) Analyze the limitations of standard approaches to 3D Elasticity when applied to edge cases in 2.002. Construct explicit counterexamples showing where intuitive reasoning fails.
2. (3 points) Prove the fundamental theorem concerning Plasticity in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
3. (6 points) Construct an original proof-level problem involving Fracture for 2.002 (Mechanics and Materials II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
4. (5 points) Analyze the limitations of standard approaches to Fatigue when applied to edge cases in 2.002. Construct explicit counterexamples showing where intuitive reasoning fails.
5. (5 points) Prove the fundamental theorem concerning Viscoelasticity in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
6. (3 points) Construct an original proof-level problem involving Composites for 2.002 (Mechanics and Materials II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
7. (3 points) Prove the fundamental theorem concerning 3D Elasticity in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
8. (4 points) Prove the fundamental theorem concerning Plasticity in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

9. (8 points) Prove the fundamental theorem concerning Fracture in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
10. (4 points) Prove the fundamental theorem concerning Fatigue in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
11. (2 points) Analyze the limitations of standard approaches to Viscoelasticity when applied to edge cases in 2.002. Construct explicit counterexamples showing where intuitive reasoning fails.
12. (4 points) Construct an original proof-level problem involving Composites for 2.002 (Mechanics and Materials II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
13. (3 points) Prove the fundamental theorem concerning 3D Elasticity in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
14. (5 points) Analyze the limitations of standard approaches to Plasticity when applied to edge cases in 2.002. Construct explicit counterexamples showing where intuitive reasoning fails.
15. (2 points) Prove the fundamental theorem concerning Fracture in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
16. (4 points) Analyze the limitations of standard approaches to Fatigue when applied to edge cases in 2.002. Construct explicit counterexamples showing where intuitive reasoning fails.
17. (6 points) Construct an original proof-level problem involving Viscoelasticity for 2.002 (Mechanics and Materials II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
18. (3 points) Analyze the limitations of standard approaches to Composites when applied to edge cases in 2.002. Construct explicit counterexamples showing where intuitive reasoning fails.
19. (4 points) Analyze the limitations of standard approaches to 3D Elasticity when applied to edge cases in 2.002. Construct explicit counterexamples showing where intuitive reasoning fails.

20. (3 points) Prove the fundamental theorem concerning Plasticity in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
21. (6 points) Prove the fundamental theorem concerning Fracture in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
22. (3 points) Construct an original proof-level problem involving Fatigue for 2.002 (Mechanics and Materials II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
23. (3 points) Prove the fundamental theorem concerning Viscoelasticity in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
24. (3 points) Construct an original proof-level problem involving Composites for 2.002 (Mechanics and Materials II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
25. (3 points) Prove the fundamental theorem concerning 3D Elasticity in the context of 2.002 (Mechanics and Materials II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

Solutions

Solution 1:

The edge case analysis reveals the subtle assumptions underlying standard treatments of 3D Elasticity. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 2:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 3:

Solution approach: First recall the definitions and key theorems related to Fracture from 2.002. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 4:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Fatigue. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 5:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 6:

Solution approach: First recall the definitions and key theorems related to Composites from 2.002. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 7:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 8:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 9:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 10:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 11:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Viscoelasticity. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 12:

Solution approach: First recall the definitions and key theorems related to Composites from 2.002. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 13:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 14:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Plasticity. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 15:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 16:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Fatigue. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 17:

Solution approach: First recall the definitions and key theorems related to Viscoelasticity from 2.002. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 18:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Composites. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 19:

The edge case analysis reveals the subtle assumptions underlying standard treatments of 3D Elasticity. By constructing explicit counterexamples, we see that the usual

hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 20:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 21:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 22:

Solution approach: First recall the definitions and key theorems related to Fatigue from 2.002. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 23:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 24:

Solution approach: First recall the definitions and key theorems related to Composites from 2.002. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 25:

The proof follows from the definitions and properties established in 2.002. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.